

Numerical Approximation of Derivatives

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1 The basic idea

A computer cannot “understand” or evaluate the writing $f'(x)$ to denote the first derivative of a function $f(x)$ at the point x .¹ How do we instruct it to produce at least a reasonably accurate numerical approximation to it?

Mathematically, the first derivative is defined as

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}. \quad (1.1)$$

The definition is unambiguous provided the limit has the same value however h approaches 0, e.g. through positive values, negative values, or jumping between positive and negative values in any way.² The computer cannot calculate the actual limit, as there is a finite smallest number that it can deal with. But we may hope that, if we take h small enough, we can find a decent approximation to $f'(x)$. How to test this hypothesis?

A reasonable expectation is that, if we take h “small enough” without passing to the limit we will have a “reasonable approximation” to $f'(x)$:

$$f'(x) \simeq \frac{f(x+h) - f(x)}{h}. \quad (1.2)$$

We can re-write this as

$$f(x+h) \simeq f(x) + hf'(x). \quad (1.3)$$

Upon comparing with the exact formula

$$f(x+h) = f(x) + \int_x^{x+h} f'(\xi) d\xi, \quad (1.4)$$

we see that (1.3) is equivalent to approximating the integral by a rectangle having base h and height $f'(x)$. How good is this approximation? Can we control the error? These are obviously crucial questions for numerical computation.

2 The Taylor theorem

The key to answering these questions is provided by *Taylor’s theorem*, one of the most useful ideas in analysis: If the function $f(x)$ has k derivatives at a point x then there is a function $R_k(x, h)$, the remainder, such that

$$f(x+h) = f(x) + hf'(x) + \frac{1}{2!}h^2f''(x) + \frac{1}{3!}h^3f'''(x) + \dots + \frac{1}{k!}h^kf^{(k)}(x) + h^kR_k(x, h). \quad (1.5)$$

and

$$\lim_{h \rightarrow 0} R_k(x, h) = 0, \quad (1.6)$$

¹There are now software packages (such as MATHEMATICA) which can deal with derivatives in the proper sense of mathematical analysis. This software can, for example, find the analytic solution of a differential equation when the equation is solvable. Since, however, the vast majority of problems are not solvable analytically, one has to recur to numerical computation and this is what we are referring to here.

²For example, if $f(x) = |x|$ and we consider $x = 0$ taking $h = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$ the limit has the value 1, but if we take $h = -1, -\frac{1}{2}, -\frac{1}{3}, -\frac{1}{4}, \dots$ the limit would be -1, and if we were to take $h = 1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \dots$ it would oscillate between ± 1 . Thus, the function $|x|$ is not differentiable at $x = 0$ (although it possesses one-sided derivatives at $x = 0$).

so that the error incurred by stopping at the k -th derivative decreases *faster* than h^k . If $f^{(k+1)}$ exists, it can be shown that

$$R_k(x, h) = \frac{h}{(k+1)!} f^{(k+1)}(x + \theta h), \quad 0 \leq \theta \leq 1. \quad (1.7)$$

In general, however, θ , which depends on both x and h , is unknown. Note that the result in (1.5) and (1.7) is true even if one stops the sum before the maximum k . So, for example,

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x + \theta_1 h), \quad f(x+h) = f(x) + hf'(x) + \frac{1}{2!} h^2 f''(x) + \frac{1}{3!} h^3 f'''(x + \theta_2 h), \quad (1.8)$$

and so on.

If the function has an infinity of derivatives in a neighborhood of x , the sum can be extended all the way to infinity to form *Taylor's series*:

$$f(x+h) = \sum_{n=0}^{\infty} \frac{h^n}{n!} f^{(n)}(x) = f(x) + hf'(x) + \frac{1}{2!} h^2 f''(x) + \frac{1}{3!} h^3 f'''(x) + \dots \quad (1.9)$$

This series may or may not converge $f(x+h)$. If it does, it is said that the function is *analytic* near x .³

Intuitively, we may say that the Taylor series is the result of the attempt to fit a polynomial of higher and higher order to f in a neighborhood of x . We would start by fitting the value at x (the first term $f(x)$), then the value of the first derivative of f , then of the second one, and so forth; see <https://www.youtube.com/watch?v=3d6DsJIBzJ4&t=520s> for a nice illustration of the idea. In particular, if we approximate $f(x)$ by fitting a straight line tangent to f at the point x and calculate the derivative of this straight line, we find (1.2).

the approximation (1.2)

2.1 Examples

1. For what kind of functions f do the formulae (1.25) and (1.26) give an exact result for the derivative? To answer this question we note from (1.24) that the function f must have zero second (and higher) derivative, i.e., it has to be a linear function, e.g. $f(x) = ax + b$. Indeed, in this case (1.25) gives

$$\frac{f(x+h) - f(x)}{h} = \frac{[a(x+h) + b] - (ax + b)}{h} = a, \quad (1.10)$$

which is the exact result.

Let us apply the formula to $f(x) = x^2$:

$$\frac{f(x+h) - f(x)}{h} = \frac{(x+h)^2 - x^2}{h} = 2x + h. \quad (1.11)$$

The first term is the exact derivative, the second one the error, which we verify here to be of first order in h , i.e., proportional to h raised to the first power.

2. A very important Taylor series is that for the function $f(x) = 1/(1-x)$. We have

$$f'(x) = \frac{1}{(1-x)^2}, \quad f''(x) = \frac{2}{(1-x)^3}, \quad (1.12)$$

etc. so that

$$\frac{1}{1-(x+h)} = \frac{1}{1-x} + \frac{h}{(1-x)^2} + \frac{1}{2} \frac{2h^2}{(1-x)^3} + \dots \quad (1.13)$$

³It should be noted that infinite differentiability is not sufficient. For example, the function $\exp(-1/x^2)$ is infinitely differentiable at $x = 0$, but all its derivatives vanish so that the sum of the Taylor series is also 0, while f does not vanish in a neighborhood of $x = 0$.

In particular, with $x = 0$, this gives

$$\frac{1}{1-h} = 1 + h + h^2 + \dots \quad (1.14)$$

which is known as the *geometric series*.

This series is quite useful in many applications. Suppose for example that we want to convert a temperature from $^{\circ}\text{F}$ to $^{\circ}\text{C}$. The exact result is found in this way:

$$T(^{\circ}\text{C}) = \frac{T(^{\circ}\text{F}) - 32}{1.8}. \quad (1.15)$$

We rewrite this as

$$T(^{\circ}\text{C}) = \frac{T(^{\circ}\text{F}) - 32}{2 - 0.2} = \frac{T(^{\circ}\text{F}) - 32}{2(1 - 0.1)} = \frac{1}{2}[T(^{\circ}\text{F}) - 32] \frac{1}{1 - 0.1} \quad (1.16)$$

Now, from (1.14), $1/(1 - 0.1) \simeq 1 + 0.1$ so that

$$T(^{\circ}\text{C}) \simeq \frac{1}{2}[T(^{\circ}\text{F}) - 32] \times 1.1. \quad (1.17)$$

In words: to convert from $^{\circ}\text{F}$ to $^{\circ}\text{C}$ subtract 32, divide by 2 and add 10%. For example, for $T = 92^{\circ}\text{F}$, by applying this rule we find $T = 33^{\circ}\text{C}$, while the exact result is 33.33°C . The same trick works if we want to convert a weight W in kg to lb:

$$W(\text{lb}) = \frac{W(\text{kg})}{0.454} \simeq \frac{W(\text{kg})}{0.45} = \frac{W(\text{kg})}{0.5 - 0.05} = \frac{W(\text{kg})}{0.5(1 - 0.1)} \simeq 2W(\text{kg}) \times 1.1K. \quad (1.18)$$

Again, double and add 10%.

3. The geometric series is a special case of the more general *binomial series*:

$$(1 + h)^{\beta} \simeq 1 + \beta h + \dots \quad (1.19)$$

for any real, positive or negative, β . If $\beta = 2$ or 3 we recover the leading terms of the expansions of $(1 + h)^2$ and $(1 + h)^3$.

4. As a final example let us take $f(x) = \cos \alpha x$. Then

$$f'(x) = -\alpha \sin \alpha x, \quad f''(x) = -\alpha^2 \cos \alpha x. \quad (1.20)$$

Then

$$\cos[\alpha(x + h)] = \cos \alpha x + h(-\alpha \sin \alpha x) + \frac{1}{2!}h^2(-\alpha^2 \cos \alpha x) + \dots \quad (1.21)$$

Let us apply this at the point $x = 0$. Then we find

$$\cos \alpha h \simeq 1 - \frac{1}{2!}h^2\alpha^2. \quad (1.22)$$

We see that this amounts to approximating the cosine function near the origin by a parabola. It is evident that the accuracy of the approximation will be very strongly dependent on the magnitude of α . A small α may give a reasonable result even with a fairly large h , while a large α will give unforgiving if h is not small enough. How good is this approximation to calculate the first point where $\cos \alpha x$ vanishes? The parabola in (1.22) vanishes for $h = \sqrt{2}/\alpha$ while $\cos \alpha h$ vanishes for $h = \pi/2\alpha$. How big is the error? Let's see:

$$\frac{\pi/2\alpha - \sqrt{2}/\alpha}{\pi/2\alpha} = \frac{\pi/2 - \sqrt{2}}{\pi/2} = 0.0997 \simeq 10\%. \quad (1.23)$$

Not too bad!

2.2 Back to $f'(x)$

By using the Taylor formula in (1.2) we find

$$\frac{f(x+h) - f(x)}{h} = \frac{f(x) + hf'(x) + \frac{1}{2!}h^2f''(x) + \frac{1}{3!}h^3f'''(x) + \dots - f(x)}{h} = f'(x) + \frac{1}{2}hf''(x) + \frac{1}{3!}h^2f'''(x) + \dots \quad (1.24)$$

This result supports our original expectation that, if h is taken small enough, (1.2) will give a decent approximation to $f'(x)$. Indeed, for h small, all terms in the right-hand side other than $f'(x)$ will be small, and they will become smaller and smaller with decreasing h . How small is small depends, of course, on the function f . If $f''(x)$ is large, for example, h must be taken very small so that the product $hf''(x)$ is not so large as to destroy the desired numerical accuracy of the formula. The function f may change slowly in a certain range of values of x , where f'' is therefore small and one can use a reasonably large value of h , but the same function may vary rapidly in a different x -interval and, there, the formula must be used with a smaller h . With these caveats, we write

$$f'(x) \simeq \frac{f(x+h) - f(x)}{h}. \quad (1.25)$$

If the increment is chosen negative, let us write it as $-h$ for clarity; then we have

$$f'(x) \simeq \frac{f(x+(-h)) - f(x)}{(-h)} = \frac{f(x) - f(x-h)}{h}. \quad (1.26)$$

Proceeding as for (1.24) we find

$$\frac{f(x) - f(x-h)}{h} = f'(x) - \frac{1}{2}hf''(x) + \frac{1}{3!}h^2f'''(x) + \dots \quad (1.27)$$

The formula (1.25) is the *forward* approximations to the first derivative since it uses a point ahead of x . Similarly, (1.26) is a *backward* approximation, since it uses a point behind x . It is evident from (1.24) that they are both affected by an error of order $O(h)$ and, therefore, they are *first-order accurate*. The difference between the true derivative and its approximation is a typical example of *truncation error*.

We note that the right-hand side of (1.25) may also be written as

$$\frac{f(x+h) - f((x+h)-h)}{h}, \quad (1.28)$$

which is exactly what we would write if we were to approximate $f'(x+h)$ by the backward formula (1.26). Thus, we may conclude that (1.25) approximates both $f'(x)$ and $f'(x+h)$ with the same error of order h . This can be better understood if we use (1.7) for the remainder with $k=0$:

$$f(x+h) = f(x) + hf'(x+\theta h), \quad (1.29)$$

from which

$$\frac{f(x+h) - f(x)}{h} = f'(x+\theta h). \quad (1.30)$$

While this would be an exact relation if the exact value of θ (which, of course, will depend both on the function f and the point x) was known, in general all we can say is that the fraction in the left-hand side gives the exact derivative at some unknown point between x and $x+h$. At any other point between x and $x+h$ all we can expect that it will approximate the derivative there with an error of maximum order h .

An obvious question now is: how can we decrease the $O(h)$ error with which $f'(x)$ is estimated? By inspecting (1.24) and (1.27) we see that

$$\frac{1}{2} \left(\frac{f(x+h) - f(x)}{h} + \frac{f(x) - f(x-h)}{h} \right) = \frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{1}{3}h^2f'''(x). \quad (1.31)$$

By using the three points x and $x \pm h$ we have succeeded in reducing the error from $O(h)$ to $O(h^2)$. How can we find a systematic way to derive high-order formulae in general without having to be so “smart” as to see this by inspection?

3 General method for the approximation of derivatives

We can see a path toward the systematic derivation of derivative approximations if we recast the derivation of the simple forward formula (1.25) as the following problem: To find constants α and β such that

$$\frac{\alpha f(x+h) + \beta f(x)}{h} \simeq f'(x) \quad (3.1)$$

with as small an error as possible. To solve this problem we expand $f(x+h)$ in a Taylor series to find

$$\frac{\alpha f(x+h) + \beta f(x)}{h} \simeq \frac{\alpha[f(x) + hf'(x) + \frac{1}{2}h^2 f''(x) + \dots] + \beta f(x)}{h} = \frac{\alpha + \beta}{h} f(x) + \frac{\alpha h}{h} f'(x) + \frac{\alpha h^2}{2h} f''(x) + \dots \quad (3.2)$$

If the result is to be as close as possible to $f'(x)$ it is evidently necessary to choose

$$\alpha + \beta = 0, \quad \alpha = 1. \quad (3.3)$$

Solving this linear system we find $\alpha = 1$, $\beta = -1$ and we recover (1.25). Evidently we must live with the error embodied in the last term of (3.2): there is nothing we can do about it because we have “spent” our two degrees of freedom α and β to satisfy the necessary relations (3.3).

The conclusion of this little analysis is that, if we want to cancel the f'' term, we need to introduce another degree of freedom. The way to do it is to invoke a third point, in addition to x and $x+h$, to approximate $f'(x)$. For example, let's try

$$\begin{aligned} \frac{\alpha f(x+h) + \beta f(x) + \gamma f(x-h)}{h} &= \frac{1}{h} \left\{ \alpha[f(x) + hf'(x) + \frac{1}{2}h^2 f''(x) + \frac{1}{3!}h^3 f'''(x) + \dots] + \beta f(x) \right. \\ &\quad \left. + \gamma[f(x) + (-h)f'(x) + \frac{1}{2}(-h)^2 f''(x) + \frac{1}{3!}(-h)^3 f'''(x) + \dots] \right\} \\ &= \frac{\alpha + \beta + \gamma}{h} f(x) + \frac{\alpha h - \gamma h}{h} f'(x) + \frac{\alpha h^2 + \gamma h^2}{2h} f''(x) + \frac{\alpha h^3 - \gamma h^3}{3!h} f'''(x) + \dots \end{aligned} \quad (3.4)$$

To approximate the first derivative we must evidently require

$$\alpha + \beta + \gamma = 0, \quad \alpha - \gamma = 1, \quad \alpha + \gamma = 0. \quad (3.5)$$

The result is

$$\alpha = \frac{1}{2}, \quad \beta = 0, \quad \gamma = -\frac{1}{2}, \quad (3.6)$$

so that

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} + \frac{h^2}{12} f'''(x) + \dots \quad (3.7)$$

The fraction in this formula is the *centered difference approximation* to the first derivative.⁴

⁴A cute comment on this result is the following: Suppose we want to apply this formula to approximating the derivative of f at the point $x + \frac{1}{2}$ by using $\frac{1}{2}h$ in place of h . We would write

$$f'(x + \frac{1}{2}h) \simeq \frac{f(x + \frac{1}{2}h + \frac{1}{2}h) - f(x + \frac{1}{2}h - \frac{1}{2}h)}{2(\frac{1}{2}h)} = \frac{f(x+h) - f(x)}{h}.$$

This shows that our relatively crude approximation (1.2) gives $f'(x)$ with an error proportional to h at any point of the interval between x and $x+h$, but it gives a smaller error proportional to h^2 at the midpoint, as shown by (3.7)! Otherwise said, the formula (1.2) approximates f' at the mid-point $x + \frac{1}{2}$ much better than at either x or $x+h$.